

Tomas Pereira - Data Scientist

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WORK EXPERIENCE

Atlas Analytics Lab

May. 2023– Dec 2024

Master's/NSERC Researcher

Concordia University, Montreal, QC

- Research and Development of a Foundational Model for Breast and Colon Cancer Histopathology Tissue
- Research and Development of a Multimodal Vision-Language Model for Automatic Report Generation using the CLIP Framework
- Recipient of the NSERC-USRA Research Award in May 2023

Concordia University

Sept. 2023 - Present

Teaching Assistant

Montreal, QC

- Teaching Assistant for the Machine Learning Course for both Graduate and Undergraduate Students
- Planned and Held Weekly Labs Teaching Theory, Pytorch, Numpy and Scikit-Learn

Bureau Veritas

July. 2022 – May. 2023

Systems Specialist

Montreal, QC

- Performed Data Wrangling and Cleaning Tasks to Assist Operation of Lab Analysts
- Refactoring and Maintenance of Datasheets and VBA Macros used in Daily Operations
- Key Figure in Development and Adoption of New Inventory System

EDUCATION

Concordia University

Sept. 2023 - Present

Master of Computer Science (MAppCompSc)

Montreal, QC

- Specialization in AI, LLMs, Reactive Programming, and Client-Server Applications
- Studied Representation Learning and Reinforcement Learning at MILA - Quebec AI Institute
- Recipient of the Concordia Merit Scholarship Award (2024 and 2025)

Concordia University

Sept 2019 - May 2023

Bachelor of Computer Science (BCompSc)

Montreal, QC

- Recipient of the Government of Quebec Award under the Merit Scholarship Program (2021)

Projects

Research Project - Updating Method-Level Comments Using Generative AI

- Fine Tuned the CodeT5 and Gemma (LLM) Models for Comment Generation and Updating.
- Studied the Effectiveness of Varied Inputs and Gradual Information Gain with Common Similarity Benchmarks

Research Project - Reinforcement Learning with Multi Instance Learning For Efficient WSI Analysis

- Developed an Agent to Reduce the Computational Load of MIL Inference on Whole Slide Images Using RL

PLAY Framework Reactive System

- Developed a Reactive Application Interfacing the YoutubeV3 API using Asynchronous Message Passing

Shipping Carrier Recommendation System

- Trained a Multiclass Classifier to be Used in a Logistics Company for Expediting Truck Selection. Led to Significant Reduction of Time Spent by dispatch.

SKILLS & INTERESTS

Skills:

- o Languages: Python, Java, SQL, VBA, C++, Ruby, C#
- o Technologies: Pytorch, Numpy, SKLearn, OpenAI API, HuggingFace, PLAY, SpeechBrain, OpenCV, Gymnasium, Pandas, SLURM
- o Other: Deep Learning, Neural Networks, R&D, Computer Vision, NLP, LLMs, Decision-Making Systems, Medical Imaging, Data Science, Backend Programming
- o Spoken Languages: English, French and Portuguese (Trilingual)
- **Interests:** Conversational AI; Dungeons and Dragons; Game Development; Gardening; Running; Rugby;